

MATH 161/162, Autumn 2025/Winter 2026
SCRIPT 6: Construction of the Real Numbers

In this sheet we define the real numbers, $\mathbb{R}$.

Definition 6.1. A subset A of $\mathbb{Q}$ is said to be a *cut* (or *Dedekind cut*) if it satisfies the following:

- (a) $A \neq \emptyset$ and $A \neq \mathbb{Q}$.
- (b) If $r \in A$ and $s \in \mathbb{Q}$ satisfies $s < r$, then $s \in A$.
- (c) A does not have a last point; i.e., if $r \in A$ then there is some $s \in A$ with $s > r$.

We denote the collection of all cuts by $\mathbb{R}$.

Lemma 6.2. Let A be a Dedekind cut and $x \in \mathbb{Q}$. Then $x \notin A$ if, and only if, x is an upper bound for A .

Proof of Lemma 6.2. Let A be a Dedekind cut and let $x \in \mathbb{Q}$. We want to prove that $x \notin A$ if and only if x is an upper bound for A .

Firstly, suppose that $x \notin A$. Therefore, by **Definition 6.1(c)**, we may say that $\forall a \in A, x > a$ as by the definition of Dedekind $\forall r \in A, \exists s \in A$ such that $s > r$. Then, as $\forall a \in A, x > a$ we may state that $\forall a \in A, x \geq a$. Thus, we may state that x is an upper bound for A if $x \notin A$.

Secondly, suppose for the sake of contradiction that $x \in A$. We want to show that x cannot be an upper bound for A . By **Definition 6.1(c)** we may say that as $x \in A, \exists s \in A$ such that $s > x$ as A has no last point. Therefore, as there exists some arbitrary value in A that is greater than x , x cannot be an upper bound.

This works to prove that if $x \notin A$ if and only if x is an upper bound for A . □

Corollary 6.3. Let A be a Dedekind cut. Then A is bounded above in $\mathbb{Q}$, but is not bounded below.

Proof. Let A be a Dedekind cut. By definition, $A \subset \mathbb{Q}$ and $A \neq \mathbb{Q}$. Hence, there exists some $q \in \mathbb{Q} \setminus A$. Since all elements of A are less than q , q is an upper bound for A in $\mathbb{Q}$.

Next, suppose A were bounded below by some $r \in \mathbb{Q}$. By the definition of a Dedekind cut, for any $x \in A$ there exists $y \in A$ with $y < x$. Therefore, for every candidate lower bound r , there is some $y \in A$ with $y < r$, contradicting that r is a lower bound. Hence, A is not bounded below in $\mathbb{Q}$.

This works to prove that the Dedekind cut A is bounded above in $\mathbb{Q}$ but not bounded below. □

Exercise 6.4. (a) Prove that, for any $q \in \mathbb{Q}$, the set

$$D_q = \{x \in \mathbb{Q} \mid x < q\}$$

is a Dedekind cut. We call D_q a *rational cut*. We also define $\mathbf{0} = D_0$.

(b) Prove that $\{x \in \mathbb{Q} \mid x \leq 0\}$ is not a Dedekind cut.

(c) Prove that $\{x \in \mathbb{Q} \mid x < 0\} \cup \{x \in \mathbb{Q} \mid x^2 < 2\}$ is a Dedekind cut.

Exercise 6.4(a). We want to show that $q \in \mathbb{Q}$, the set $D_q = \{x \in \mathbb{Q} \mid x < q\}$ is a Dedekind cut. For this, it suffices to show that $D_q = \{x \in \mathbb{Q} \mid x < q\}$ complies with the definition of a Dedekind cut.

Firstly, we want to show that $D_q \neq \emptyset$ or $D_q \neq \mathbb{Q}$. By the definition of the set we know that it contains at least 1 element and therefore $D_q \neq \emptyset$. Next, suppose there is an arbitrary $q \in \mathbb{Q}$. By the definition of $\mathbb{Q}$ we know there exists some $q' \in \mathbb{Q}$ such that $q' > q$.

Secondly, we want to show that if $r \in D_q$ and $s \in \mathbb{Q}$ such that $s < r$ then $s \in D_q$. Suppose we have an arbitrary $q \in D_q$. Let there exists a $q_2 \in \mathbb{Q}$ such that $q_2 < q$. As $q_2 < q$ and $D_q \{x \in \mathbb{Q} \mid x < q\}$ we may say that $q_2 \in D_q$. As this applies for arbitrary values of q and q_2 , this works as proof for being downwards closed.

Thirdly, we want to show that D_q has no last point. Suppose for contradiction that there exists a greatest element $r \in D_q$. By the definition of D_q , we have $r < q$. Since the rationals are dense, there exists a rational number $r' \in \mathbb{Q}$ such that $r < r' < q$. Then $r' \in D_q$ and $r' > r$, which contradicts the assumption that r is the greatest element. Therefore, D_q has no last point. $\square$

Exercise 6.4(b). We want to prove that $\{x \in \mathbb{Q} \mid x \leq 0\}$ is not a Dedekind cut. For this, it suffices to show that it contains a last point.

By the definition of the subset, we may say that $0 \in \{x \in \mathbb{Q} \mid x \leq 0\}$. Now, we want to show that $\{x \in \mathbb{Q} \mid x \leq 0\}$ does not contain any value greater than 0.

As $\{x \in \mathbb{Q} \mid x \leq 0\}$ is bounded below, we may say that $\forall x \in \{x \in \mathbb{Q} \mid x \leq 0\}, x \leq 0$. As 0 is the largest value, there does not exist some $s \in A$ such that $s > r$ and hence 0 is the largest value. Therefore, A has a last point of 0 as all values in $\{x \in \mathbb{Q} \mid x \leq 0\}$ are lesser than 0. $\square$

Exercise, 6.4(c). Let

$$D = \{x \in \mathbb{Q} \mid x < 0\} \cup \{x \in \mathbb{Q} \mid x^2 < 2\}.$$

We want to show that D is a Dedekind cut.

First, D is nonempty because $0 \in D$ since $0^2 = 0 < 2$. It is not all of $\mathbb{Q}$ because, for example, $2 \notin D$, as $2 > 0$ and $2^2 = 4 > 2$.

Next, D is downward closed. Suppose $x \in D$ and $y \in \mathbb{Q}$ with $y < x$. If $x < 0$, then $y < 0$, so $y \in D$. If $x \geq 0$ and $x^2 < 2$, then $y < x$ implies that $y^2 < x^2 < 2$ whenever $y \geq 0$, or $y < 0$ in which case $y \in D$ anyway. In either case, $y \in D$.

Finally, D has no greatest element. If $x < 0$, then taking $z = x/2$ gives $x < z < 0$, so $z \in D$ and $z > x$. If $x \geq 0$ and $x^2 < 2$, then because the rationals are dense, there exists a rational z with $x < z$ and $z^2 < 2$, so again $z \in D$ and $z > x$. Thus, no element of D can be the greatest.

Since D is nonempty, downward closed, and has no greatest element, it satisfies all the conditions of a Dedekind cut.

Hence D is a Dedekind cut. □

Definition 6.5. If $A, B \in \mathbb{R}$, we say that $A < B$ if A is a proper subset of B . (Recall from Script 1 that A is a proper subset of B if $A \subset B$ but $A \neq B$.)

Exercise 6.6. Show that $\mathbb{R}$ satisfies Axioms 1,2 and 3.

Lemma 6.7. *A nonempty subset of $\mathbb{R}$ that is bounded above has a supremum.*

Proof of 6.7. Let $A \subset \mathbb{R}$ be a nonempty set of Dedekind cuts that is bounded above. We want to show A has a supremum.

As A is bounded above, there exists a Dedekind cut B such that

$$a \subseteq B \quad \text{for all } a \in A.$$

Define

$$\gamma = \bigcup_{a \in A} a.$$

Because A is nonempty, γ is nonempty. Moreover, since each $a \in A$ is contained in B , we have

$$\gamma \subseteq B \subsetneq \mathbb{Q},$$

so we may say that γ is a proper subset of $\mathbb{Q}$.

If $q \in \gamma$ and $r \in \mathbb{Q}$ with $r < q$, then $q \in a$ for some $a \in A$. Since a is downward closed, it follows that $r \in a \subseteq \gamma$, showing that γ is downward closed. Next, if $q \in \gamma$, then $q \in a$ for some $a \in A$. Since a has no greatest element, there exists $r \in a$ such that $q < r$, and hence $r \in \gamma$. Thus γ has no greatest element. Therefore γ is a Dedekind cut.

By construction, $a \subseteq \gamma$ for all $a \in A$, so γ is an upper bound of A . If δ is any upper bound of A , then $a \subseteq \delta$ for all $a \in A$, which implies

$$\gamma = \bigcup_{a \in A} a \subseteq \delta.$$

Hence, this works to prove that γ is the least upper bound of A . □

Exercise 6.8. Show that $\mathbb{R}$ satisfies Axiom 4. (*Hint: Additional Exercise 4 of Script 5 is very useful here.*)

Proof. We want to show that $\mathbb{R}$ satisfies Axiom 4. We will prove this through contradiction. Suppose that $\mathbb{R}$ does *not* satisfy Axiom 4. Then there exist nonempty, disjoint, open sets $A, B \subset \mathbb{R}$ such that

$$A \cup B = \mathbb{R}.$$

Since A and B are nonempty, choose $a \in A$ and $b \in B$. Without loss of generality, assume $a < b$. Because A and B are disjoint, $a \neq b$.

Now, we can define the set:

$$X := A \cap [a, b].$$

Then X is nonempty (since $a \in X$) and bounded above by b . By completeness of $\mathbb{R}$, the supremum

$$s := \sup X$$

exists and satisfies $a \leq s \leq b$.

Since $A \cup B = \mathbb{R}$ and $A \cap B = \emptyset$, we must have either $s \in A$ or $s \in B$. Therefore, we consider these two cases separately.

Case 1: $s \in A$.

Because A is open, there exists $\varepsilon > 0$ such that

$$(s - \varepsilon, s + \varepsilon) \subset A.$$

Since s is the supremum of X , there exists $x \in X$ with

$$s < x < s + \varepsilon.$$

But then $x \in A$ and $x \leq b$, so $x \in X$, contradicting that s is an upper bound of X .

Case 2: $s \in B$.

Because B is open, there exists $\varepsilon > 0$ such that

$$(s - \varepsilon, s + \varepsilon) \subset B.$$

Since $s = \sup X$, there exists $x \in X$ with

$$s - \varepsilon < x \leq s.$$

Then $x \in A$ by definition of X , and $x \in B$, contradicting that A and B are disjoint.

As both cases lead to contradictions. Therefore, $\mathbb{R}$ satisfies Axiom 4. This works to prove that $\mathbb{R}$ satisfies Axiom 4. $\square$

Is there only one continuum satisfying Axioms 1–4? We have constructed one such continuum, but it turns out there are others. Consequently, we add one more axiom. There is only one continuum that satisfies all five axioms, and as we'll see, it's $\mathbb{R}$.

Definition 6.9. Let C be a continuum satisfying Axioms 1-4. Consider a subset $X \subset C$. We say that X is *dense* in C if every $a \in C$ is a limit point of X .

Lemma 6.10. Let X be a subset of C . Then the following are equivalent:

- a) X is dense in C
- b) $\overline{X} = C$
- c) Given $a, b \in C$, there is some $x \in X$ such that $a < x < b$.

Proof of Lemma 6.10. Let X be a subset of C . We want to show that the three statements are equivalent.

Firstly, we show that X is dense in C is equivalent to $\overline{X} = C$. Suppose X is dense in C . By definition, every point $c \in C$ is either in X or is a limit point of X . Therefore, by the definition of closure, $\overline{X} = X \cup LP(X) = C$. Conversely, suppose that $\overline{X} = C$. Then, every point $c \in C$ satisfies $c \in X$ or $c \in LP(X)$. In other words, every point of C is either in X or arbitrarily close to points of X , which is exactly the definition of X being dense in C . This establishes the equivalence of (a) and (b).

Next, we show that $\overline{X} = C$ is equivalent to that there exists some $x \in X$ such that $a < x < b$. Suppose $\overline{X} = C$. Let $a, b \in C$ with $a < b$. Since $b \in C = \overline{X}$, it is either in X or a limit point of X . In either case, there exists $x \in X$ such that $a < x < b$, because if $b \in X$, we can pick $x = b - \epsilon$ for sufficiently small $\epsilon > 0$, and if b is a limit point, there must exist points of X arbitrarily close to b from below. Conversely, suppose that for every $a < b$ in C there exists $x \in X$ with $a < x < b$. Then, for any $c \in C$ and any interval $(c - \epsilon, c + \epsilon)$, there exists $x \in X$ within that interval. This shows that every $c \in C$ is either in X or is a limit point of X , so $\overline{X} = C$. This establishes the equivalence of (b) and (c).

Combining the two equivalences, we conclude that all three statements (a), (b), and (c) are equivalent. $\square$

Our next goal is to prove that $\mathbb{Q}$ is dense in $\mathbb{R}$. Just to make sense of that statement, we need to decide how to think of $\mathbb{Q}$ as a subset of $\mathbb{R}$. For every rational number $q \in \mathbb{Q}$, define the corresponding real number as the Dedekind cut

$$i(q) = D_q = \{x \in \mathbb{Q} \mid x < q\}.$$

For example, $0 = i(0)$. Check that this gives a well-defined injective function $i: \mathbb{Q} \rightarrow \mathbb{R}$. We identify $\mathbb{Q}$ with its image $i(\mathbb{Q}) \subset \mathbb{R}$ so that the rational numbers $\mathbb{Q}$ are a subset of the real numbers $\mathbb{R}$. (Similarly $\mathbb{N}$ and $\mathbb{Z}$ can also be understood as subsets of $\mathbb{R}$.)

Theorem 6.11. $i(\mathbb{Q})$ is dense in $\mathbb{R}$.

Proof. We want to show that $i(\mathbb{Q})$ is dense in $\mathbb{R}$. By definition, this means that between any two real numbers there exists a rational number. Therefore, it suffices to prove that for any $A, B \in \mathbb{R}$ with $A < B$, there exists a rational $q \in \mathbb{Q}$ such that

$$A < i(q) < B.$$

Since $A < B$, we have $A \subset B$ as a proper subset. Therefore, there exists some $p \in B \setminus A$. Because B is a Dedekind cut and has no last point, there exists $p' \in B$ such that $p' > p$. Consider the cut $i(p') \subset \mathbb{Q}$. By construction, $i(p') \subset B$ and $i(p') \neq B$, so we have $i(p') < B$. Moreover, for all $a \in A$, we have $a < p < p'$, which implies $a \in i(p')$. Also, $p \in i(p')$, so $A < i(p')$.

Thus, we have found $i(p') \in \mathbb{Q}$ such that

$$A < i(p') < B,$$

showing that between any two real numbers there exists a rational number. This completes the proof. $\square$

We finish this section with several useful consequences of the density of the rationals in the reals.

Corollary 6.12 (The Archimedean Property). *Let $A \in \mathbb{R}$ be a positive real number. Then there exist nonzero natural numbers $m, n \in \mathbb{N}$ such that $i(\frac{1}{n}) < A < i(m)$.*

Similarly, if $A \in \mathbb{R}$ is a negative real number, then there exist nonzero natural numbers $m, n \in \mathbb{N}$ such that $i(-m) < A < i(-\frac{1}{n})$.

Proof. Suppose there exists $A \in \mathbb{R}$ such that $A > 0$. By **Theorem 6.11**, there exists a rational number $p \in \mathbb{Q}$ such that

$$0 < p < A.$$

Now, we may write $p = \frac{a}{b}$ where $a, b \in \mathbb{N}$. Then, as $a \in \mathbb{N}$ we may say that $a \geq 1$. Then, we have that

$$\frac{1}{b} \leq \frac{a}{b} \text{ and } p < A.$$

By setting b and n , we obtain the lower bound

$$i\left(\frac{1}{n}\right) = i\left(\frac{1}{b}\right) \leq p < A,$$

so $i(1/n)$ provides a concrete positive rational number strictly less than A , which establishes the desired lower bound.

Next, since $\mathbb{R}$ has no largest element, there exists $B \in \mathbb{R}$ with $A < B$. Again, by **Theorem 6.11**, there exists a rational number $q \in \mathbb{Q}$ such that

$$A < q < B.$$

Now suppose $q = \frac{c}{d}$ where $c, d \in \mathbb{N}$. Since c is a positive integer, we have $c \geq 1$. Because $q = \frac{c}{d}$ and $d \geq 1$, it follows that

$$q = \frac{c}{d} \leq c.$$

Then, as we have the inequality $A < q$ we may combine these inequalities to get

$$A < q \leq c.$$

Then, by assigning c as m , we then obtain a natural number m such that

$$A < i(m),$$

which provides the desired upper bound for A .

Lastly, by combining the two inequalities, we have that

$$i\left(\frac{1}{n}\right) < A < i(m),$$

The case $A < 0$ follows by applying the same reasoning to $-A > 0$. Hence, the other proof is analogous.

This works to prove the Archimedean Principle. □

Corollary 6.13. *$i(\mathbb{N})$ is an unbounded subset of $\mathbb{R}$.*

Proof. We want to show that $i(\mathbb{N})$ is an unbounded subset of $\mathbb{R}$. For this, it suffices to show that $i(\mathbb{N})$ has no upper bound in $\mathbb{R}$. Suppose, for the sake of contradiction, that there exists some real number $A \in \mathbb{R}$ which is an upper bound for $i(\mathbb{N})$; that is,

$$i(n) \leq A \quad \text{for all } n \in \mathbb{N}.$$

By the Archimedean Property, there exists a natural number $m \in \mathbb{N}$ such that

$$A < i(m).$$

But this contradicts the assumption that A is an upper bound of $i(\mathbb{N})$, since $i(m) \in i(\mathbb{N})$ is greater than A .

Hence, our assumption must be false, and there can be no real upper bound of $i(\mathbb{N})$. Therefore, $i(\mathbb{N})$ is unbounded in $\mathbb{R}$. □

Corollary 6.14. *If $A \in \mathbb{R}$ is a real number, then there is a unique integer n such that $i(n-1) \leq A < i(n)$. Similarly, there is a unique integer m such that $i(m-1) < A \leq i(m)$.*

Proof of Corollary 6.14. We want to prove that if $A \in \mathbb{R}$ then there is a unique integer n such that $i(n-1) \leq A < i(n)$ and a unique integer m such that $i(m-1) < A \leq i(m)$. We will prove this through direct proof using the Archimedean Property (**Corollary 6.13**). We prove the first statement; the second follows by analogous reasoning.

By the Archimedean Property, there exists an integer k such that $A < i(k)$. Now consider

$$S := \{n \in \mathbb{Z} \mid A < i(n)\}.$$

and therefore we may say that S is nonempty.

Let $n_0 \in S$ be the smallest integer such that $A < i(n_0)$. We may say such an integer exists as $\mathbb{Z}$ is discrete and ordered. Since n_0 is the smallest integer with this property, we must have $A \geq i(n_0-1)$. Hence we can make the statement that,

$$i(n_0-1) \leq A < i(n_0).$$

To prove uniqueness, suppose n_0 and n_1 both satisfy

$$i(n_0-1) \leq A < i(n_0) \quad \text{and} \quad i(n_1-1) \leq A < i(n_1).$$

If $n_0 \neq n_1$, assume without loss of generality that $n_0 < n_1$. Then $n_0 \leq n_1-1$, so

$$i(n_0) \leq i(n_1-1) \leq A,$$

which contradicts $A < i(n_0)$. Therefore $n_0 = n_1$, proving uniqueness.

Finally, the proof for rounding up ($i(m-1) < A \leq i(m)$) is analogous. □

Since $\mathbb{Q}$ is a countable set, Theorem 6.11 motivates one more (and last) axiom.

Axiom 5. *The continuum contains a countable dense subset.*

We just showed that $\mathbb{R}$ satisfies Axiom 5. In the rest of this sheet, we will show that the real numbers $\mathbb{R}$ are the “only” continuum. Exactly what we mean by this is specified in Theorem 6.18.

Definition 6.15. Let X and Y be sets with orderings $<_X$ and $<_Y$, respectively. A function $f: X \rightarrow Y$ is *order-preserving* if for all $r, s \in X$,

$$r <_X s \implies f(r) <_Y f(s).$$

Note that the function $i: \mathbb{Q} \rightarrow \mathbb{R}$ discussed above is order-preserving.

Exercise 6.16. Let C satisfy Axioms 1–5. Let $K \subset C$ be a countable dense subset of C . Construct an order-preserving bijection $f: \mathbb{Q} \rightarrow K$.

Exercise 6.17. Let $f: \mathbb{Q} \rightarrow K$ be an order-preserving bijection, as found in the previous exercise. Let $A \in \mathbb{R}$. Then $A \subset \mathbb{Q}$ and so $f(A) \subset K \subset C$. Define $F: \mathbb{R} \rightarrow C$ by

$$F(A) = \sup f(A).$$

(a) Show $\sup f(A)$ exists, so F is well-defined.

(b) Show F is injective and order-preserving.

Theorem 6.18. Suppose that C is a continuum satisfying Axioms 1–5. Then C is isomorphic to the real numbers $\mathbb{R}$; i.e., there is an order-preserving bijection $F: \mathbb{R} \rightarrow C$.

Theorem 6.18 says that Axioms 1–5 completely characterize the continuum, so we can say that the real numbers are the “only” continuum. As a consequence, we will no longer refer to C as anything but $\mathbb{R}$.

Additional Exercises

1. Show that if $A \in \mathbb{R}$ and $0 < A$, then $0 \in A$.
2. Let $X = \{\frac{m}{2^n} \mid m, n \in \mathbb{Z}\}$. Prove that X is dense in $\mathbb{R}$. (X is called the set of *dyadic rationals*.)
3. Define the *frontier* of a subset $X \subset \mathbb{R}$ to be the set $\partial X = \overline{X} \cap \overline{(\mathbb{R} \setminus X)}$.
 - (a) Show that ∂X is closed.
 - (b) Find examples where ∂X is finite, infinite, and empty.
 - (c) Is it true that $\partial X \supset \overline{X} \setminus X$ or $\partial X \subset \overline{X} \setminus X$?
 - (d) Can ∂X be dense in $\mathbb{R}$?
4. In this exercise, we will prove that $\mathbb{R}$ is uncountable.
 - (a) Prove that $\wp(\mathbb{N})$ is uncountable (recall that $\wp(\mathbb{N})$ is the set of all subsets of $\mathbb{N}$), if not done already in *Additional Exercise 1.12*.
 - (b) For $A \subset \mathbb{N}$ and $n \in \mathbb{N}$, let

$$C_{A,n} = \left\{ p \in \mathbb{Q} \mid p < \sum_{i \in [n] \cap A} 10^{-i} \right\}.$$

(Note that the sum above is finite.)

Prove that the set $C_A = \bigcup_{n \in \mathbb{N}} C_{A,n}$ is a Dedekind cut.

- (c) Prove that if $A, B \subset \mathbb{N}$ are such that $A \neq B$, then $C_A \neq C_B$. Conclude that $\mathbb{R}$ is uncountable.

Hint: Let m be the minimum element of the symmetric difference of A and B , suppose without loss of generality that $m \in A \setminus B$, let

$$q_m = \frac{10^{-m}}{9}$$

and prove that $q_m \in C_A \setminus C_B$.

Of course, the exercise above also works replacing 10 with any other natural that is at least 3. Replacing 10 with 2 still yields an uncountable subset of $\mathbb{R}$, but one has to be careful as there are countably many collisions: for a finite nonempty set A , we have $C_A = C_B$ where $B = (A \setminus \{\max A\}) \cup \{n \in \mathbb{N} \mid n > \max A\}$.